CUSTOMER BEHAVIOR ANALYSIS

1. Introduction

Customer Behavior Analysis helps businesses understand customer purchasing patterns, customer preferences, and retention trends. By analyzing customer data, organizations can improve customer satisfaction and make better business decisions.

2. Objective

The main objective of this project is to analyze customer behavior using transaction data. The project focuses on customer segmentation, purchase pattern analysis, and retention trend analysis to provide meaningful business recommendations.

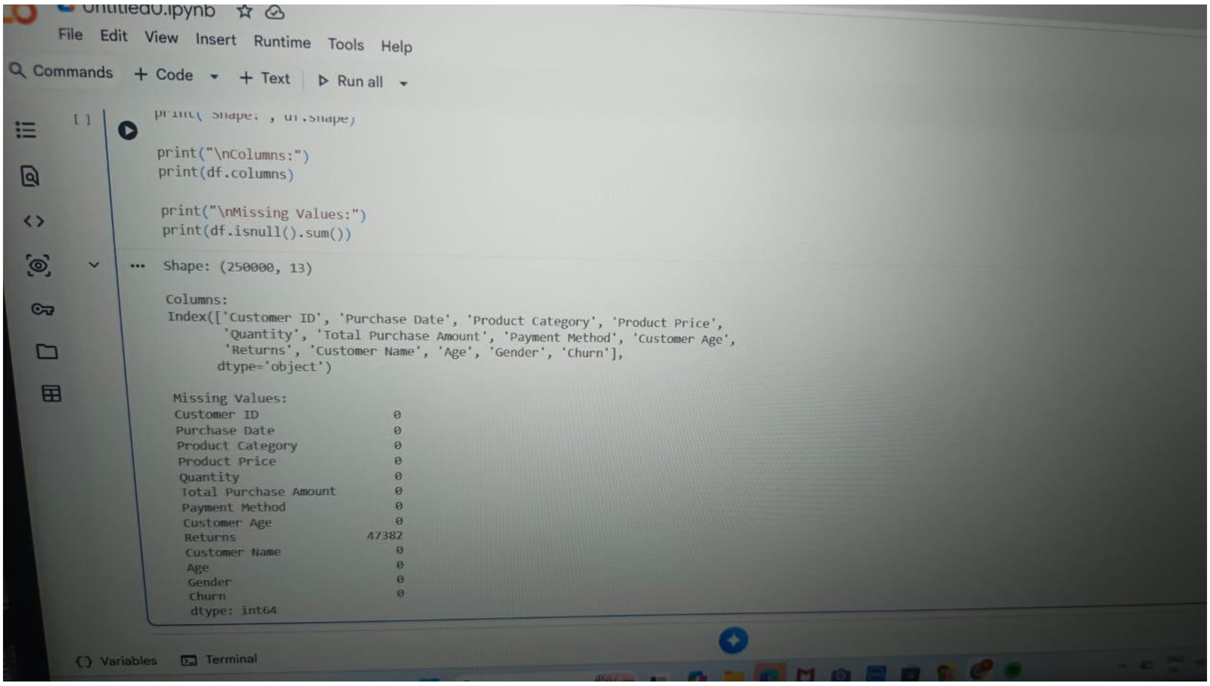
3. Dataset Description

The dataset contains customer information such as Age, Gender, Product Category, Purchase Amount, Payment Method, and Churn Status. The dataset consists of 250,000 records and 13 attributes.

4. Data Cleaning and Feature Engineering

The dataset was examined for missing values and inconsistencies. Missing values in the Returns column were identified and handled appropriately. Relevant features were selected for analysis and customer segmentation.

Screenshot 1:



5. Customer Segmentation

Customer segmentation was performed using the K-Means Clustering algorithm. Customers were grouped based on Age and Total Purchase Amount. Four customer segments were identified.

Segment Summary

Cluster	Average Age	Average Purchase Amount
0	42.82	864.18
1	43.77	3382.96
2	44.84	4610.25
3	43.80	2129.47

Interpretation

- Cluster 0 → Low Value Customers
- Cluster 1 → High Spending Customers
- Cluster 2 → Premium Customers
- Cluster 3 → Medium Value Customers

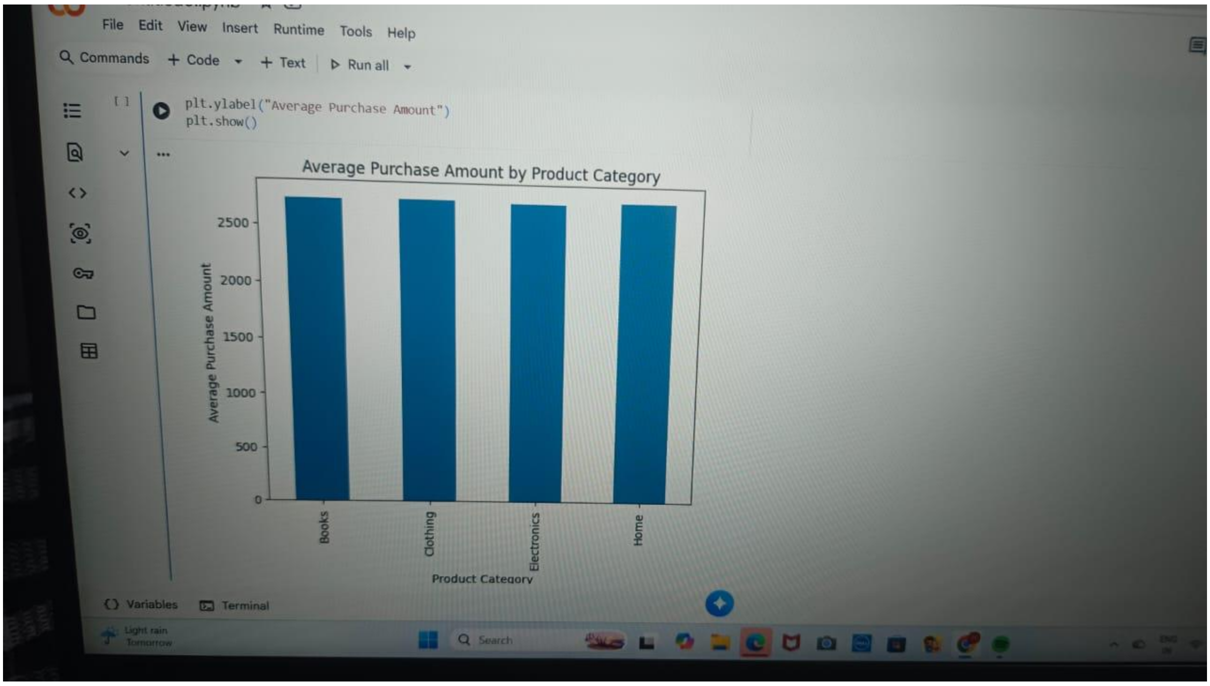
Screenshot 2:

6. Purchase Pattern Analysis

The purchase behavior of customers across different product categories was analyzed. The average purchase amount was compared among Books, Clothing, Electronics, and Home categories.

The analysis shows that customer spending is relatively balanced across all product categories.

Screenshot 3:

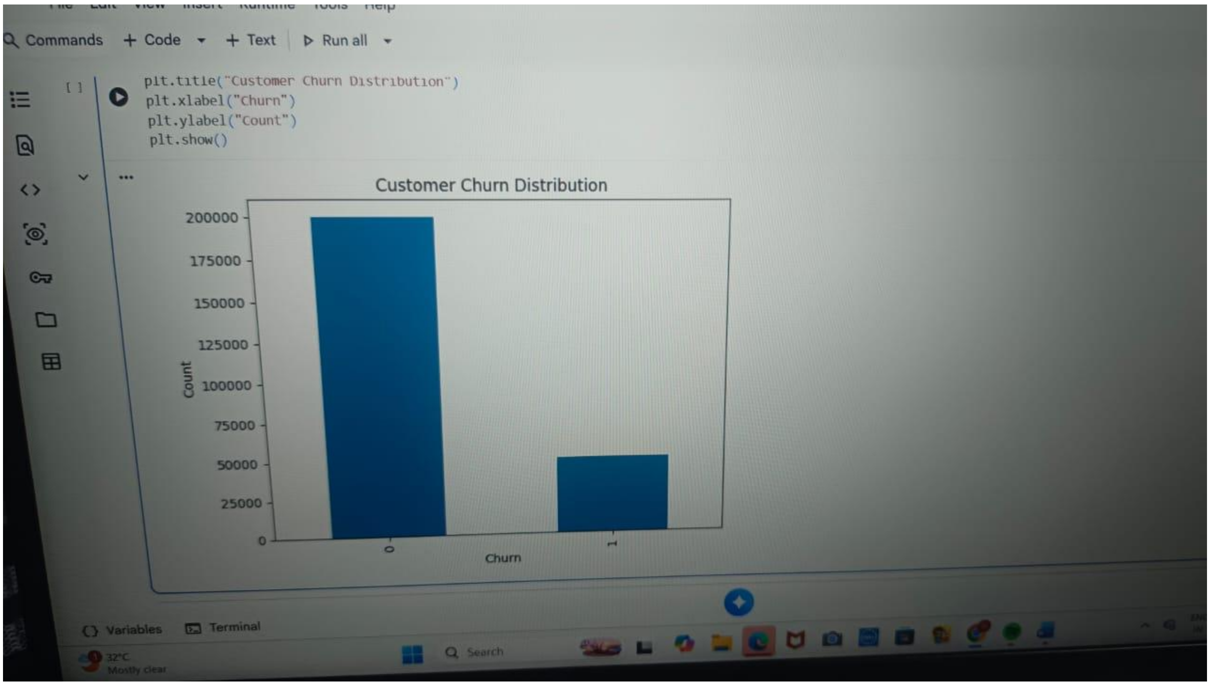


7. Customer Churn Analysis

Customer churn analysis was conducted to understand customer retention behavior.

The results indicate that approximately 80% of customers remain active, while around 20% of customers have churned.

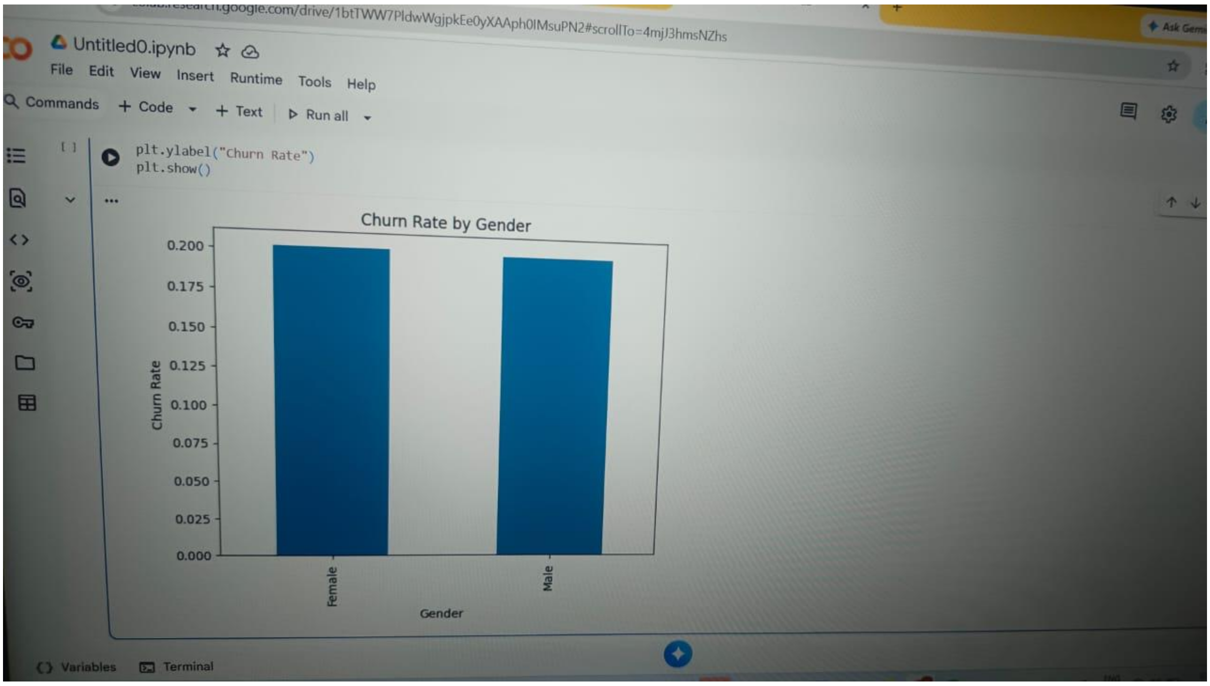
Screenshot 4:



The churn rate was also compared across genders.

The analysis shows that male and female customers exhibit similar churn behavior with only minor differences.

Screenshot 5:

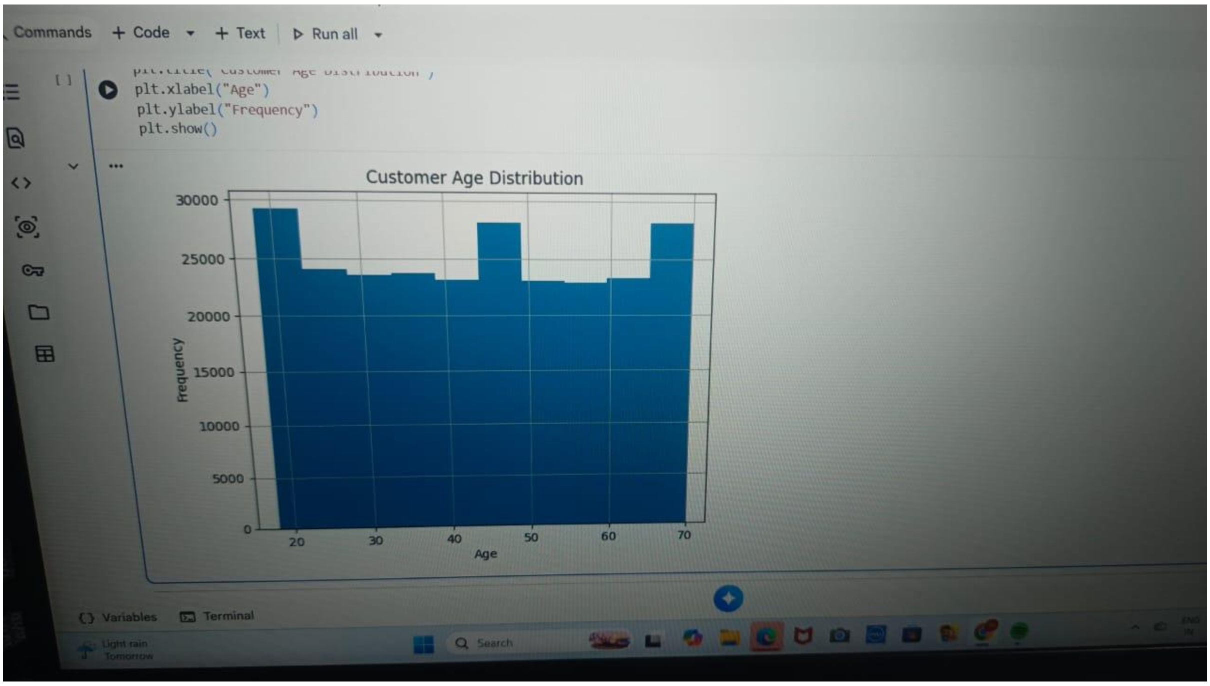


8. Customer Age Distribution

Customer age distribution was analyzed to understand the demographic spread of customers.

The graph indicates that customers belong to different age groups ranging from 18 to 70 years. The customer base is well distributed across various age categories.

Screenshot 6:



9. Findings

- ➢ The dataset contains 250,000 customer records.
- ➢ Customer churn rate is approximately 20%.
- ➢ Male and female customers show similar churn behavior.
- ➢ Product categories have nearly equal average purchase amounts.
- ➢ K-Means clustering successfully identified four customer segments.
- ➢ Customer age distribution is balanced across different age groups.

10. Recommendations

- ➢ Provide loyalty rewards to premium customers.
- ➢ Offer personalized discounts to medium-value customers.
- ➢ Target low-value customers through promotional campaigns.
- ➢ Implement email and SMS marketing strategies for customer retention.
- ➢ Introduce membership and reward programs to increase customer engagement.

11. Conclusion

This project successfully analyzed customer behavior using data analytics techniques. Customer segmentation, purchase pattern analysis, and churn analysis provided valuable insights into customer preferences and retention trends. The recommendations suggested can help businesses improve customer satisfaction, increase retention, and enhance overall business performance